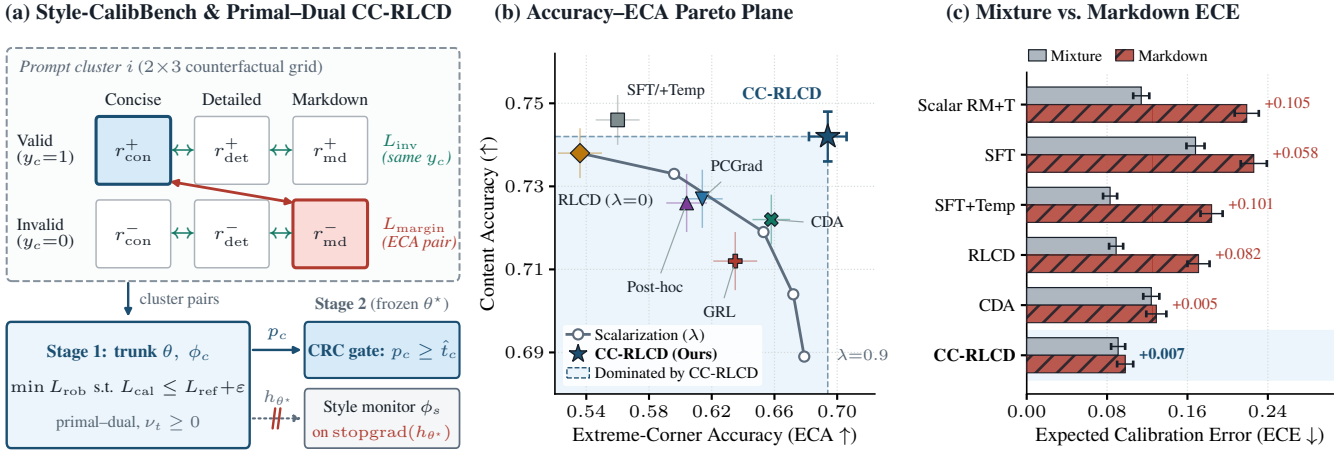

(a) Style-CalibBench & Primal–Dual CC-RLCD
Prompt cluster i (2×3 counterfactual grid)
Concise
Detailed
Markdown
Valid ($y_c=1$)
Invalid ($y_c=0$)
r^+_{con}
r^+_{det}
r^+_{md}
r^-_{con}
r^-_{det}
r^-_{md}
L_{inv} (same y_c)
L_{margin} (ECA pair)
cluster pairs
Stage 1: trunk θ, ϕ_c
$\min L_{\mathrm{rob}}$ s.t. $L_{\mathrm{cal}} \le L_{\mathrm{ref}}+\varepsilon$
primal–dual, $\nu_t \ge 0$
p_c
Stage 2 (frozen $\theta^\star$)
CRC gate: $p_c \ge \hat{t}_c$
$h_{\theta^\star}$
Style monitor ϕ_s
on stopgrad($h_{\theta^\star}$)
(b) Accuracy–ECA Pareto Plane
SFT/+Temp
CC-RLCD
PCGrad
RLCD (λ=0)
CDA
Post-hoc
GRL
Scalarization (λ)
CC-RLCD (Ours)
Dominated by CC-RLCD
λ=0.9
Content Accuracy (↑)
Extreme-Corner Accuracy (ECA ↑)
0.75
0.73
0.71
0.69
0.54
0.58
0.62
0.66
0.70
(c) Mixture vs. Markdown ECE
Mixture
Markdown
Scalar RM+T
SFT
SFT+Temp
RLCD
CDA
CC-RLCD
+0.105
+0.058
+0.101
+0.082
+0.005
+0.007
0.00
0.08
0.16
0.24
Expected Calibration Error (ECE ↓)